Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Stewpid
Project / Solution Name	Bridge - Reskilling Platform
Name (Leader)	Mahir Jabid Nafi
Email & Phone (Leader)	stewpeed.dev@gmail.com, (+60) 142411337
University Affiliation(s)	Taylor's University
Live Prototype URL	https://bridge-reskill.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/mjNafi-dev/Bridge---Reskilling-Platform

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Bridge is one web app hosted on Replit. A React app (built with Vite) is served by a Python FastAPI server, which also runs the API. When a CV is uploaded, the server reads it in memory, pulls out roles, education and skills with rules, and matches each skill to the ESCO skills list using sentence embeddings. An optional mode can send the CV text to Claude instead, but it is off in the live demo. Matching scores 117 roles from a SQLite database built from public data (ESCO, O*NET, ILO and live job-posting counts). Each score combines how much the user already knows, how fast the gap closes, future demand under two scenarios and interest fit, and each role is tagged with UN SDGs. The pathway then shows hand-checked courses for each topic, filtered by budget. Learners can share their journey with a short code, which HR users compare with a hiring brief. Universities get a growth dashboard behind an admin sign-in. The ESCO embeddings were made once on a RunPod GPU and committed, so the live server only embeds the user's own phrases, on CPU.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript / React 18.3	Frontend UI	MIT
React Router 6.28	Page routing	MIT
Vite 5.4 + @vitejs/plugin-react 4.3	Frontend build tool / dev server	MIT
jsPDF 4.2	"Save as PDF" on a shared journey: builds the PDF in the browser	MIT
Python 3.11+	Backend language	PSF License
FastAPI ≥0.115, Uvicorn ≥0.32	API server	MIT / BSD-3-Clause

Name & Version	Purpose in Project	License
NumPy ≥2.0	Score and similarity maths	BSD-3-Clause
fastembed ≥0.9 (Qdrant)	CPU embeddings for user phrases	Apache 2.0
pypdf ≥5.0	Text out of PDF CVs	BSD-3-Clause
python-docx ≥1.1	Text out of Word CVs	MIT
python-multipart ≥0.0.9	File uploads in FastAPI	Apache 2.0
Anthropic Python SDK ≥0.60	Optional AI CV reading (Claude)	MIT
openpyxl	Reading the ILO spreadsheet when refreshing trend data (offline script)	MIT
fpdf2	Generating the synthetic sample CV PDF (offline dev script, not in the app)	LGPL-3.0
sentence-transformers + PyTorch (on RunPod)	The one-off GPU embedding job	Apache 2.0 / BSD-3-Clause
Bricolage Grotesque (Google Fonts)	Heading font	SIL Open Font License 1.1
JetBrains Mono (Google Fonts)	Labels and numbers font	SIL Open Font License 1.1
Switzer (Fontshare, Indian Type Foundry)	Body text font	ITF Free Font License (free for personal and commercial use)
Simple Icons v16.35.0 (Claude, Gemini, MCP marks) and v13.21.0 (OpenAI mark)	Assistant marks in the "Learn with your own AI" demo	CC0 1.0 (marks belong to their owners)

Name & Version	Purpose in Project	License
3D icons from thiings.co (upload, folder closed/open, uploading, accepted, retry; sprout, seedling, bonsai, tree, blossom; toolbox, wrenches, hammers, screwdriver, screw, nail; AI sparkle; paper plane; clipboard; shipping box, globe, video, book, laptop, document, certificate, graduation cap; the empty toolbox, packed box and blue plane were edited by me)	CV upload state illustrations; familiarity growth stages; the Your skills toolbox and Resources box; Your AI, Share my journey and Your path cards	Free to use under thiings.co terms; credited in the README

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Anthropic Claude API via Replit AI Integrations	Optional AI mode for CV reading (structured extraction of roles, skills, education)	Sponsor-provided (Replit AI Integrations); built in but not switched on in the live demo
Open Library Search API (Internet Archive)	Book search in the onboarding interests step	Free, no key
ESCO API (European Commission)	Occupations, skills, ISCO codes (offline data build)	Free, no key
Remotive public API	Live remote job postings, counts only (offline trend refresh)	Free
Arbeitnow public API	Live job postings, counts only (offline trend refresh)	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Deployments + workspace)	Hosts the whole app: FastAPI serves the API and the built React app from one server (.replit, scripts/build.sh)	Replit Core (sponsor-provided); region Asia, 0.5 vCPU / 2 GiB RAM
RunPod GPU Pod (1× NVIDIA A40, PyTorch 2.8.0 template), pod bridge-skill-embeddings, ID f2w43seqazno9r	One-off job embedding 3,943 ESCO skills and job titles (6.23 s on GPU, ~14 min pod uptime)	Sponsor-provided US$25 credit
GitHub	Code repository	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
BAAI/bge-small-en-v1.5	Hugging Face (BAAI)	Pre-trained, no fine-tuning: ESCO skill/job embeddings (GPU on RunPod), user phrases (CPU in the API), and matching a hiring brief's skills to what a candidate shared	MIT
Claude Opus 5.5 (claude-opus-5-5)	Anthropic, via Replit AI Integrations	Pre-trained: optional CV reading into structured fields. Built in, but not switched on in the live demo	Anthropic commercial terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO occupations & skills (API v1.2)	esco.ec.europa.eu	Public	Free reuse with attribution (Commission Decision 2011/833/EU)	N
ESCO–O*NET crosswalk v1 (Sep 2022)	esco.ec.europa.eu	Public	Free reuse with attribution	N
O*NET 31.0 career interest types	onetcenter.org	Public	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET Bright Outlook occupations (31.0)	onetonline.org	Public	CC BY 4.0	N
ILO 2025 Global Index of Occupational Exposure to GenAI (Gmyrek et al., WP 140)	github.com/pgmyrek/2025_GenAI_scores_ISCO08	Public	CC BY 4.0	N
Live job posting counts (daily snapshot of ~2,900 ads; titles cleaned, matched to one role each, then discarded)	Remotive, Arbeitnow	Public	Free API terms (attribution)	N
UN Sustainable Development Goals (goal names)	sdgs.un.org/goals	Public	UN public information	N
Course catalogue: 164 courses (titles, links, level, hours, price), checked 23 Oct 2026	Compiled by the team from providers' official pages (OpenLearn, MIT OCW, Kaggle, Khan Academy, Coursera, Cisco, HubSpot and others); backend/data/courses.json	Public	Links and facts only, no course content copied	N
Sample CV (graphic designer, PDF)	Made by the team	Synthetic	n/a	N
Four demo people (designer, developer, student, technician)	Made by the team	Synthetic	n/a	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo University: 4 programmes, 126 made-up students (made-up names), 12 weeks of progress	Generated by backend/scripts/make_org_demo.py, learning-curve model; best-fit jobs from each student's strongest modules	Synthetic	n/a	N
Four example candidates + a junior UI designer brief (hiring demo)	Made by the team, backend/data/journeys_demo.json	Synthetic	n/a	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), Claude Opus 5.5	Writing the code, data scripts and documentation. Not used for the idea, problem framing or design	Major
Replit Agent	Setting up the imported project on Replit, and updating one dependency version after a security scan	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Claude Code was used only to write the code. The problem framing, the solution idea and the product decisions were mine, and I designed every screen myself in Figma before it was built. Every change was run and checked in the browser before I kept it, and I compared each feature with my Figma designs and the hackathon rules. Numbers in the app come only from cited public data, never from AI. CV reading was tested on several made-up CVs in different formats. I can explain every part of the code.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core plan	Hosting and deployment of the prototype
RunPod	RunPod credit (US$25)	One-off GPU job that embedded the ESCO skills and job titles

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts the live prototype; the app is a standard FastAPI + React build, so it could move to another host. The RunPod job produced the skill embeddings that power meaning-based matching; it ran once and its output is committed, so the app runs without a GPU, though re-running it needs one. The optional Claude CV reading is off in the live demo, and CVs are read by rules.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No accounts or personal data are stored on the server. Answers, the chosen pathway and progress stay in the user's own browser. An uploaded CV is read in memory and then thrown away; it is never saved or logged. Name, country and age range are filled in for the user to check. Email, phone and address are never read. The optional AI mode would send the CV text to Claude, but it is off in the live demo. A shared journey is saved on the server only when the learner asks for a code. It holds only what the share page shows, and only people with the code can see it. Hiring projects stay in the HR user's browser, and the brief comparison ignores name, age and country. A university sees a student's progress only if the student joins with its code and agrees to it.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Embeddings can mix up phrases that share words but not meaning (for example "unit testing" and "test electronic units"). To limit this, a skill tag needs a high match score, and every answer filled from a CV is shown for the user to check. Demand scores are estimates that combine US growth flags, a daily sample of job ads and the ILO exposure index. The app labels them as estimates, shows two scenarios and gives the source and date. The job ads come mostly from European boards, so some roles show few or no ads. The catalogue covers 117 roles, not every job. The "ahead of others" figure compares against typical workers built from ESCO data, not real people, and says so. Rule-based CV reading can miss unusual layouts. No core feature depends on an AI provider.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Date: 25 October 2026